STAFF REVIEW · H446 265 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A print spooler.

2. Replaces ACC with supplied input.

3. Blocked until an event such as I/O completes.

4. Q=0; it samples D at the clock edge.

I Do — source worked steps

1. Predict first — INP sets ACC=8; STA X writes 8 to mailbox X; the ACC remains 8. HLT ends the program.

We Do — guided CHECK

1. Predict first
The process waits for disk completion, then becomes Ready for selection by the scheduler.

You Do — source mark schemes

1. ACC=8 after INP [2]; X becomes 8 after STA [1]; ACC remains 8 [1].

2. Ready waits CPU [1], Running executes [1], I/O request [1] causes Waiting [1], I/O completion [1] returns Ready [1].

3. OS file lookup [1], permission [1], driver [1], possible Waiting [1], spool queue [1], scheduled printer completion [1].

Exit Ticket — source answer

1. False. It normally enters Waiting.